Femtojoule Pulse Reconstruction via Kerr Instability Amplification

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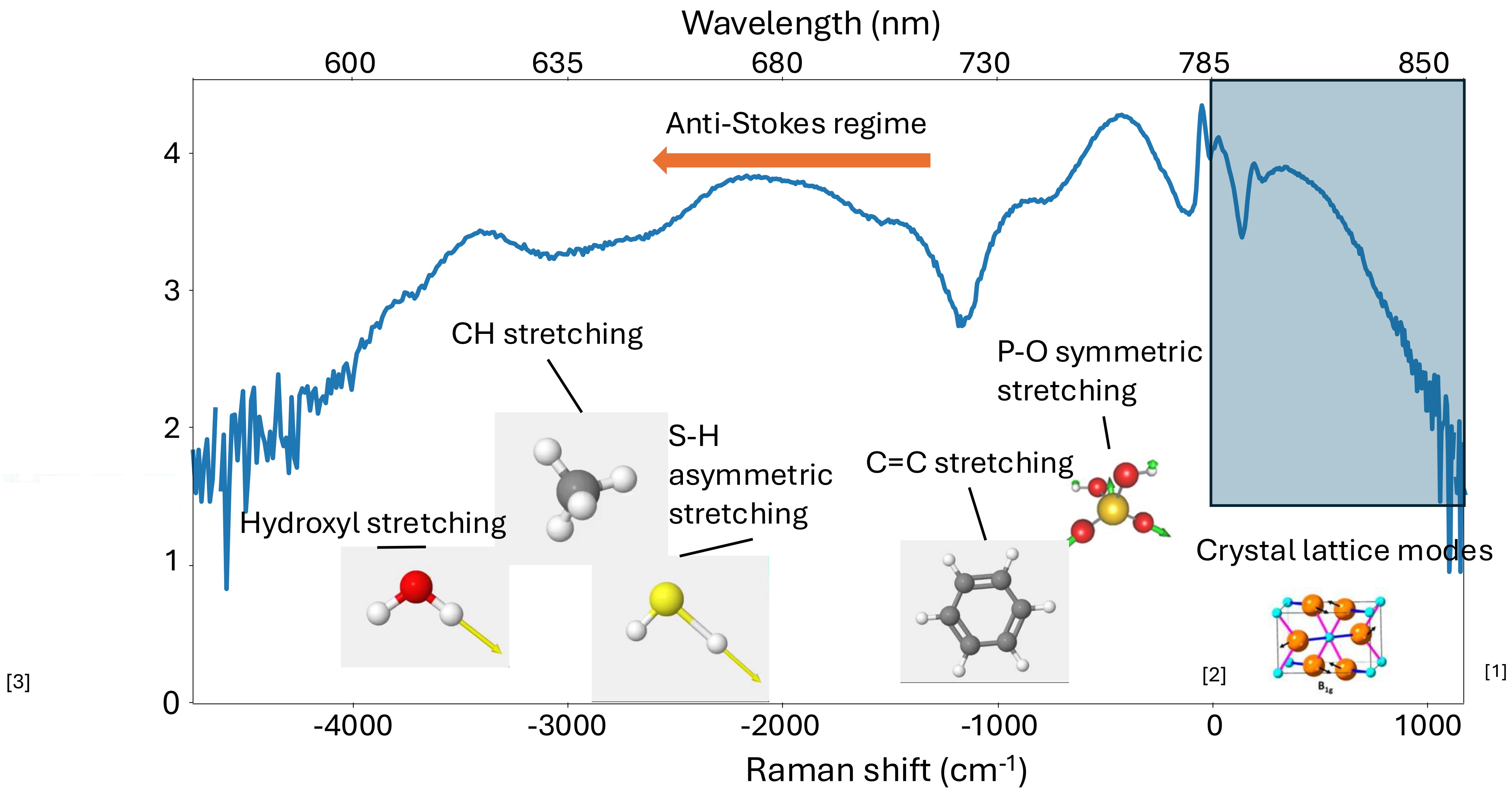
University of Windsor

Attosecond
Condensed
Matter
Experiments

ACME LAB



N. G. Drouillard and T. J. Hammond, “Stimulated Raman Spectroscopy Using a Tunable Visible Broadband Probe Pulse Generated by Kerr Instability Amplification” Applied Spectroscopy, 2025.

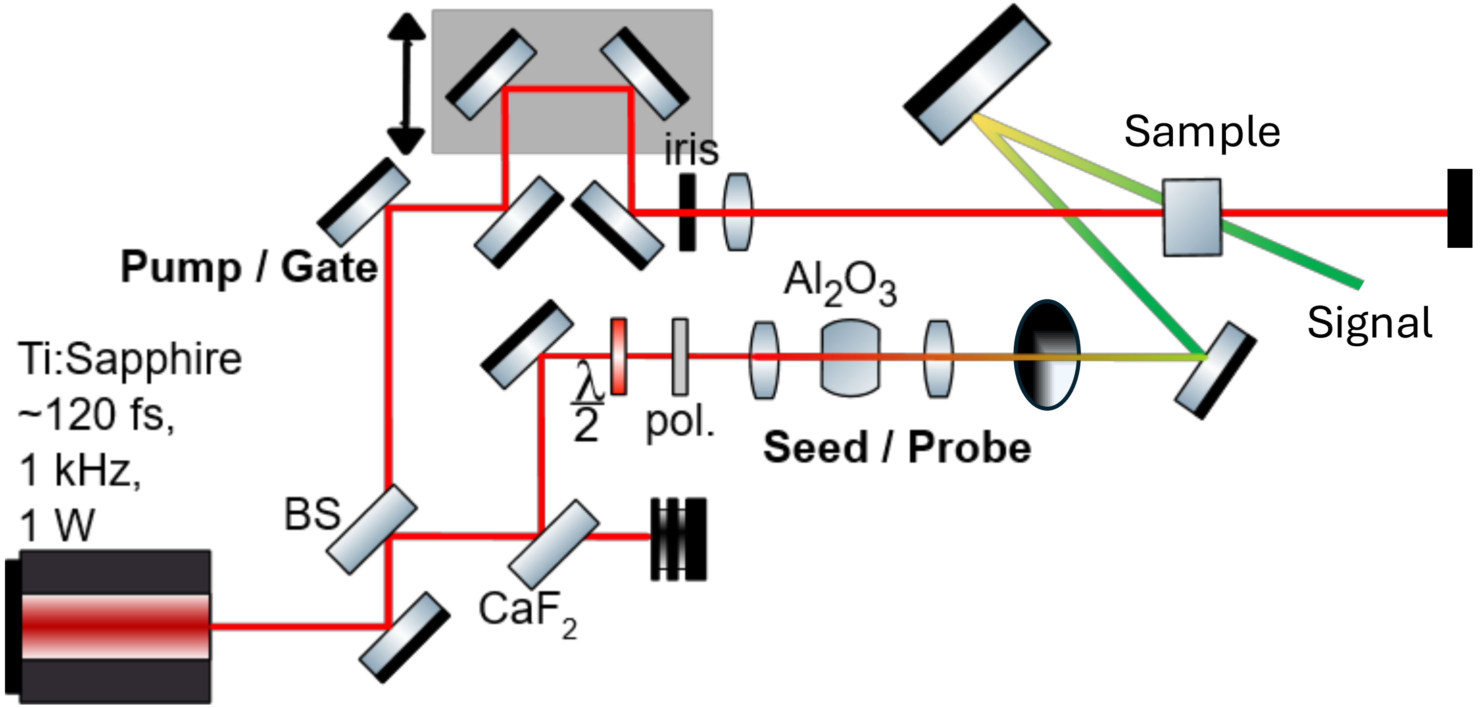
Outline

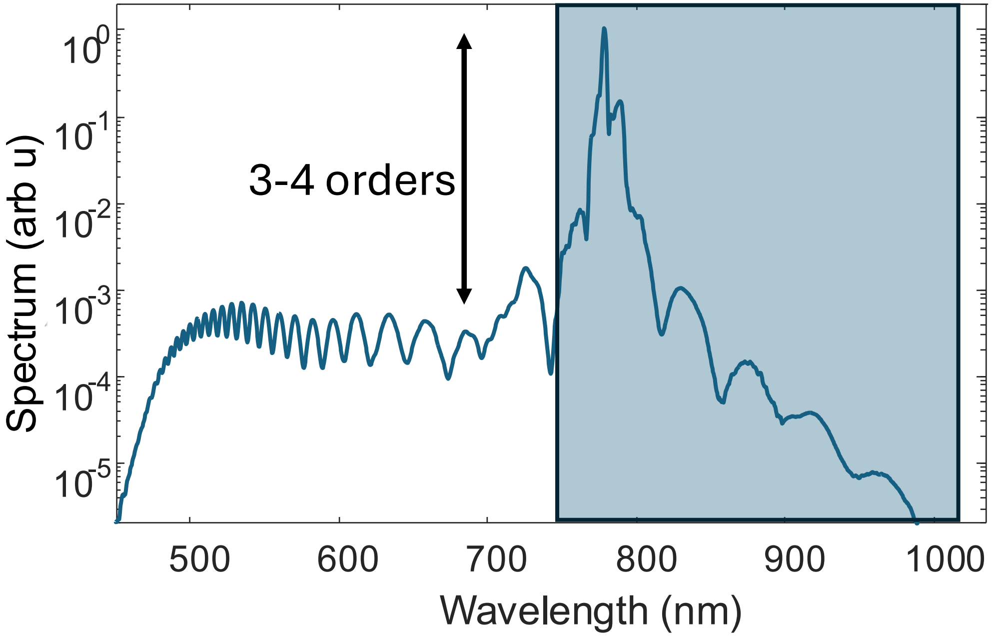
*SHG = second harmonic generation
*PG = polarization gating
*OPA = optical parametric amplification
*KIA = Kerr instability amplification

Challenge: reconstruct weak (fJ) octave-spanning pulses

- SHG FROG
  - Phase matching limits bandwidth and sensitivity
- PG XFROG
  - Poor SNR at low powers
- OPA XFROG
  - Phase matching + gain narrowing limit bandwidth
- KIA + XFROG (KICKING FROG)
  - Broadband amplification + background-free

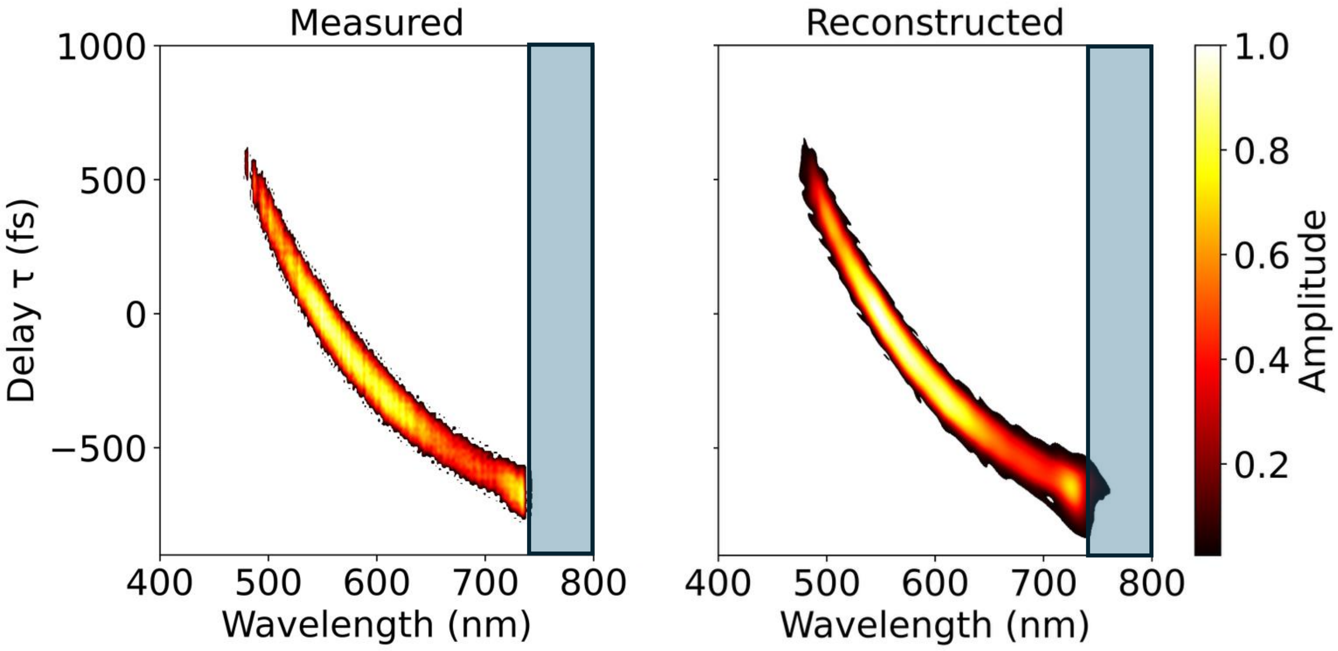
Reconstructing broadband pulses





Polarization **g**ating **c**ross-correlation **f**requency-**r**esolved **o**ptical **g**ating (PG XFROG)
Kerr-**i**nstability **c**ross-**k**orrelation **i**nduced **n**onlinear **g**eneration **FROG** (KICKING FROG).

PG XFROG reconstruction



Polarization Gating

$$G(t) = \sin\left[\frac{\pi L}{\lambda}\, n_{2B}\, I_p(t)\right] \approx \frac{\pi L}{\lambda} n_{2B} I_p(t)$$

Spectrogram

$$I_{PG-XFROG}(\omega,\tau) = \left| \int_{-\infty}^{\infty} E_p(t) G(t-\tau) e^{-i\omega t}\, dt \right|^2$$

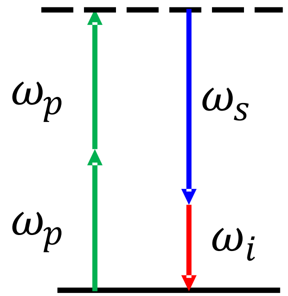
Limited in dynamic range by polarizers

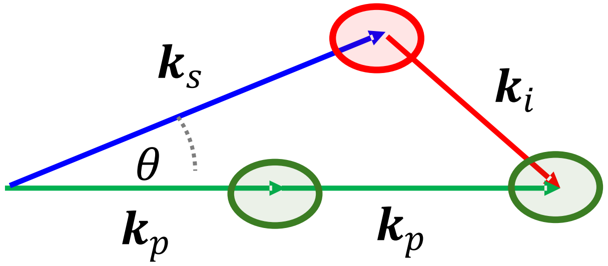
Extreme four-wave mixing

*Kerr instability amplification (**KIA**)

Conservation of energy and momentum

-phase matching



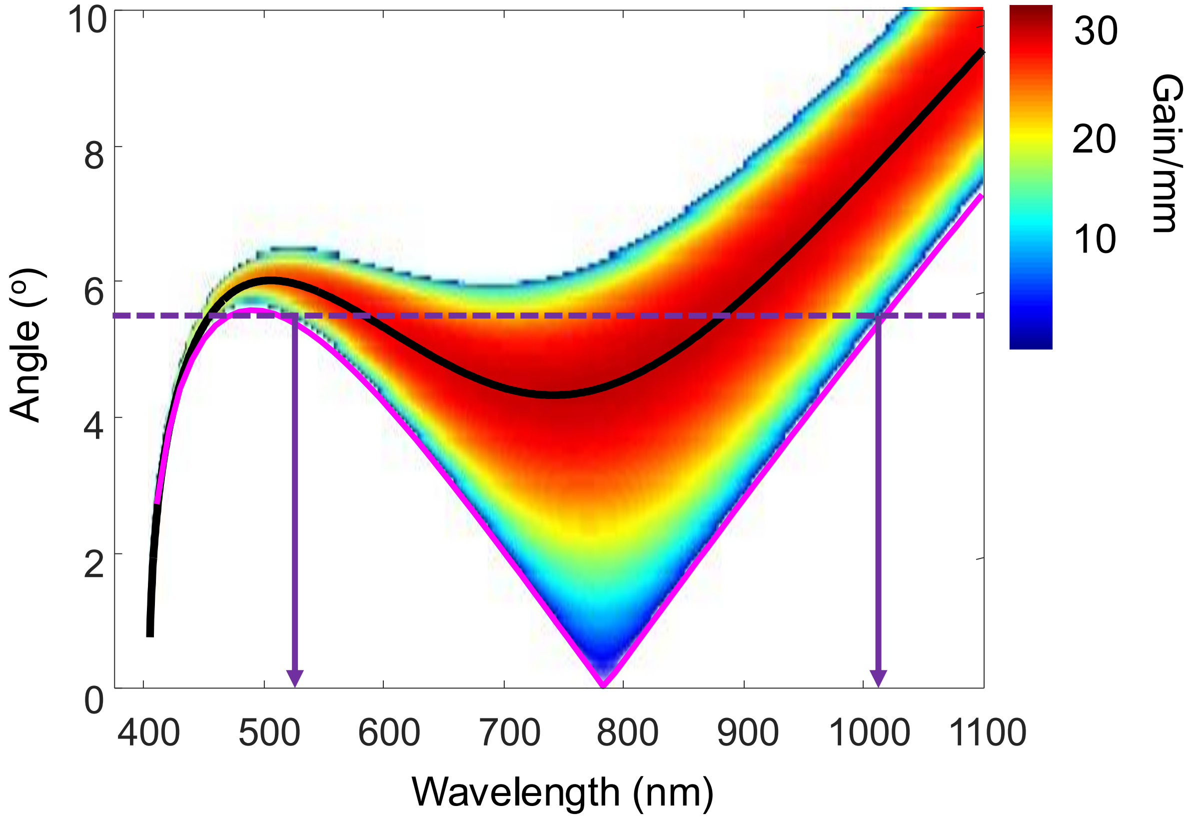


$$\Delta \boldsymbol{k} = \underbrace{\boldsymbol{k_s} + \boldsymbol{k_i} - 2\boldsymbol{k_p}}_{\text{FWM}} + \boldsymbol{k_p} n_2 I_p$$

To the extreme

Leads to gain:

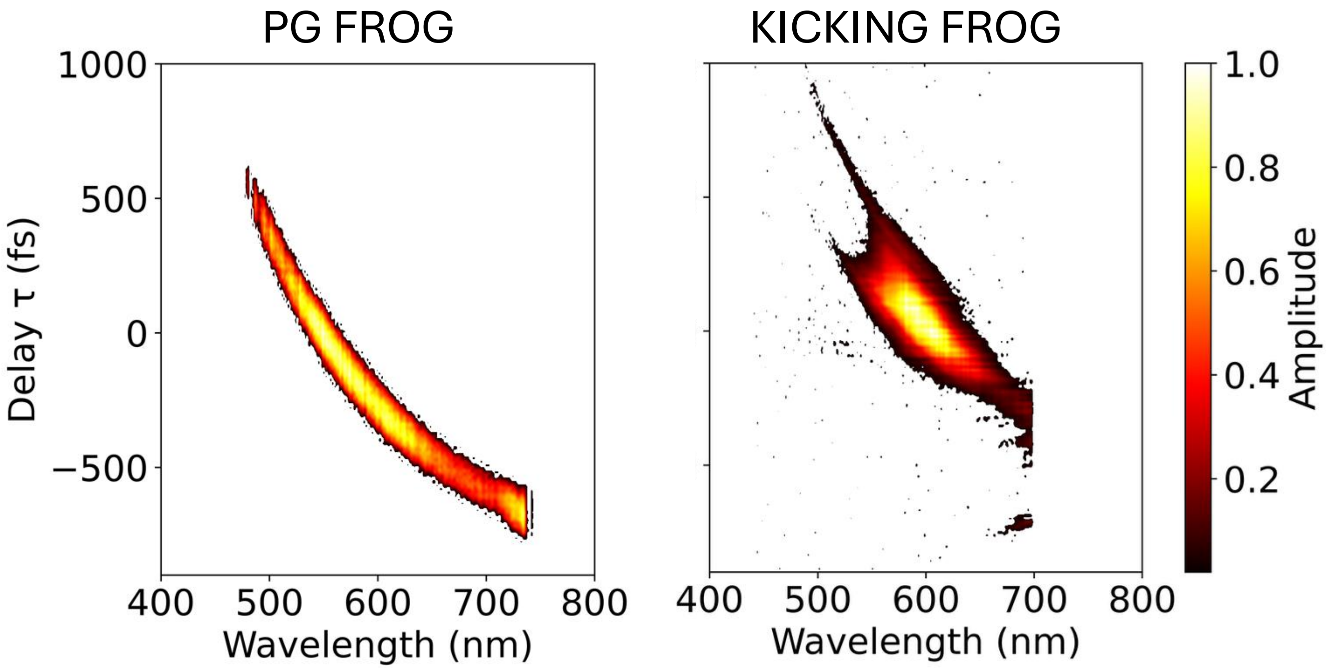
$$G = \cosh^2\left[k_p \frac{n_2 I_p L}{n_p}\right] \approx \frac{1}{4} e^{k_p \frac{n_2 I_p}{n_p} L}$$



FWM (magenta), KIA max gain (black line), full intensity-dependent gain (colourbar).

S. Ghosh, N. G. Drouillard, and T. J. Hammond, “Single-stage few-cycle pulse amplification,” *Phys. Rev. A*, 109 (2024)

KICKING FROG results

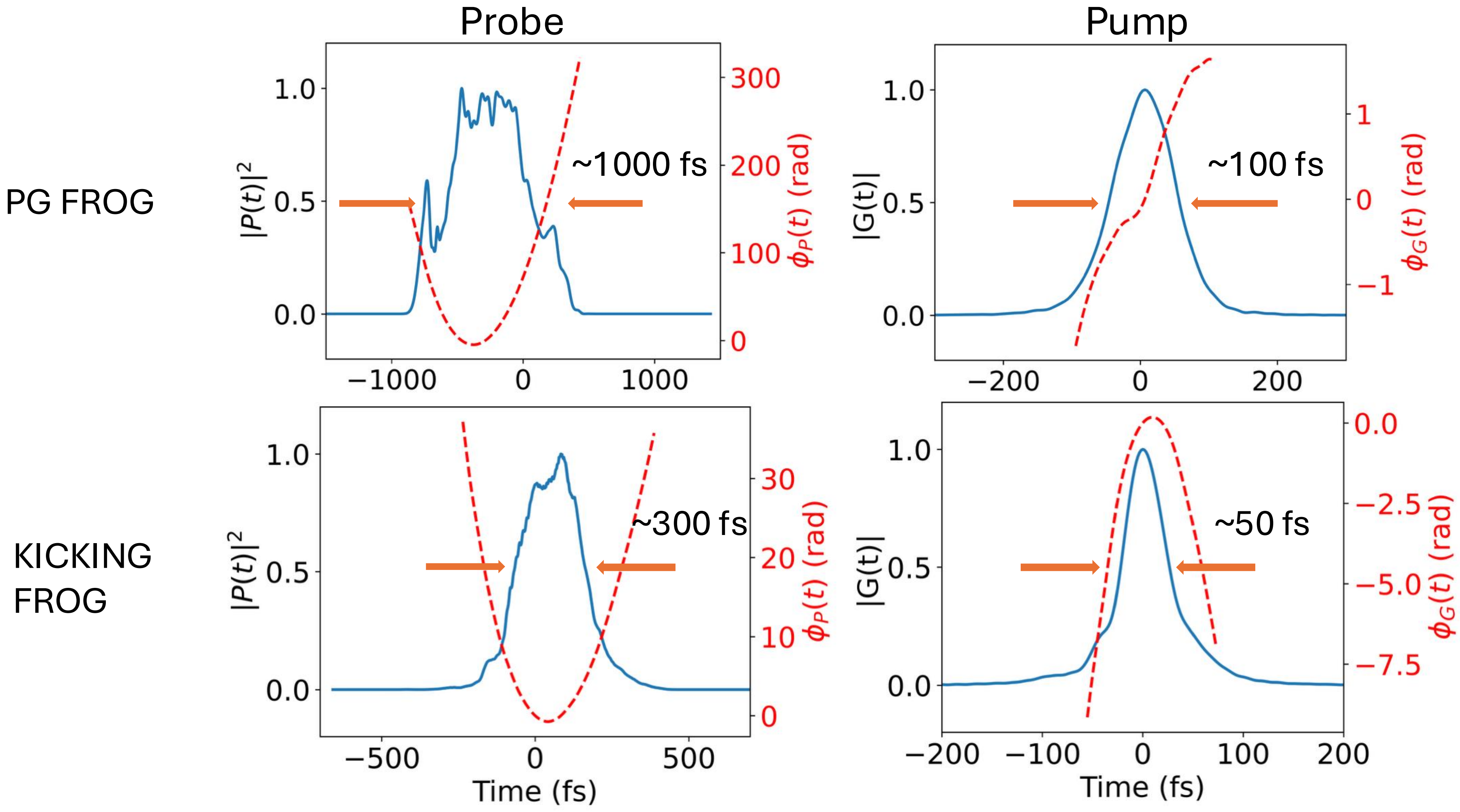


Reproduces the chirp, but also gives information on the amplification

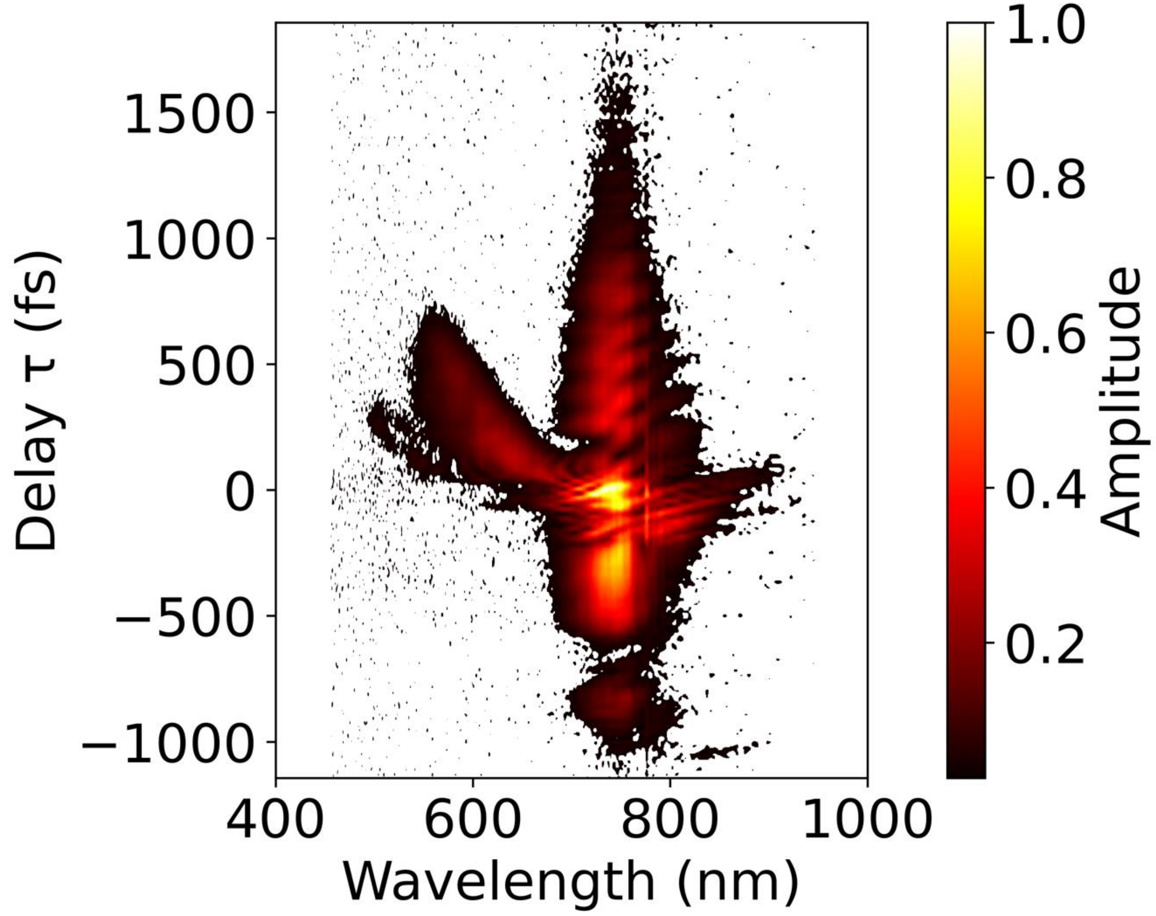
Bandwidth is limited compared to PG FROG

N. G. Drouillard et al., "Influence of Kerr anisotropy in parametric amplification," *Phys. Rev. A*, vol. 113, p. 043513 , Apr. 2026.

Reconstruction

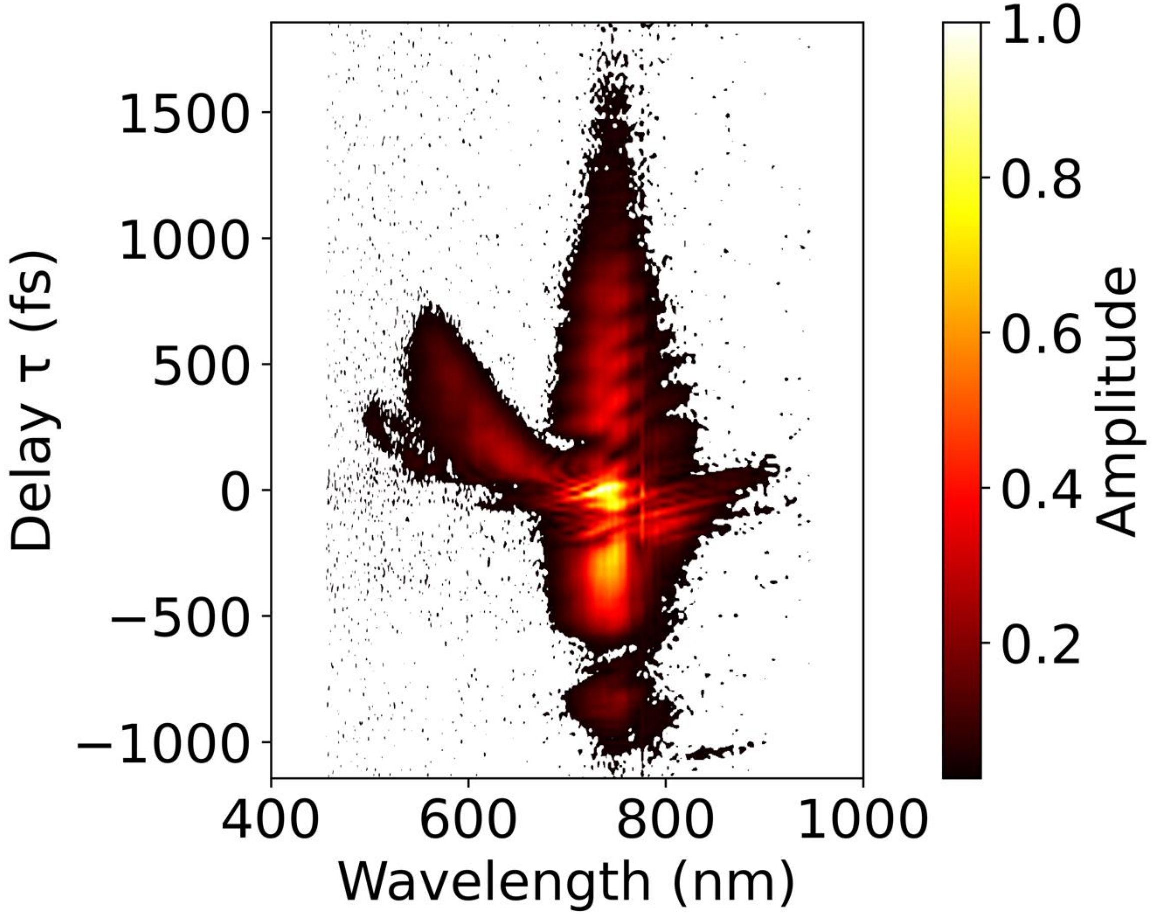


KICKING FROG results



Can have larger dynamic range
Can measure main portion of spectrum

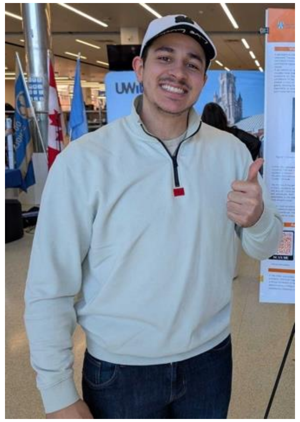
KICKING FROG results



Conclusions

- We use Kerr-instability amplification (KIA) as a novel source for Raman spectroscopy
- To reconstruct broadband and weak femtosecond pulses, KIA provides a convenient nonlinear optical signal.
- We call our method “Kerr-instability cross-korrelation induced nonlinear generation FROG” (KICKING FROG)

The ACME Lab at UWindsor



Experimental Work: Nathan Drouillard and Abdullah Mustafa

Manipal University